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PT Detective: A VR System for Reviewing Squat Biomechanics

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CSCI 1951T: Augmented Data Visualization for Research · Brown University · Spring 2026

INTRODUCTION

Physical therapists routinely assess movement quality from a single camera angle on 2D video — a flat representation that loses depth information and obscures laterally-oriented patterns. Knee valgus during squat descent, weight shifting onto one limb, and subtle hip drop are linked in clinical literature to ACL injury risk and post-rehabilitation limb deficiency [1,2,3], yet can be invisible to the naked eye when reviewed on flat video.

This work introduces **PT Detective**, a Meta Quest 3 application that captures squats via inside-out body tracking and replays them as a color-coded 3D skeleton in a virtual room, where the reviewer walks freely around the body. The system was evaluated against 2D side-view video review in a within-subjects pilot study (n = 8).

SYSTEM DESIGN

PT Detective consists of two Unity 2022.3 scenes built on the Meta Movement SDK. In the **Patient scene**, a participant performs five squats while inside-out body tracking captures per-frame transforms for fifteen joints, serialized to JSON on-device. In the **PT Review scene**, the recorded skeleton replays inside a virtual room. The reviewer walks freely around the playback, can pause and step frame-by-frame, and rotates the playback yaw with a controller stick.

An analysis pipeline computes six biomechanical metrics each frame against literature-backed thresholds (Table 1). Joints are colored by severity: gray (normal), amber (mild), red (severe). Severe joints are scaled 1.5× and outlined in white for accessibility beyond color alone. A three-frame persistence filter suppresses single-frame sensor-noise spikes.

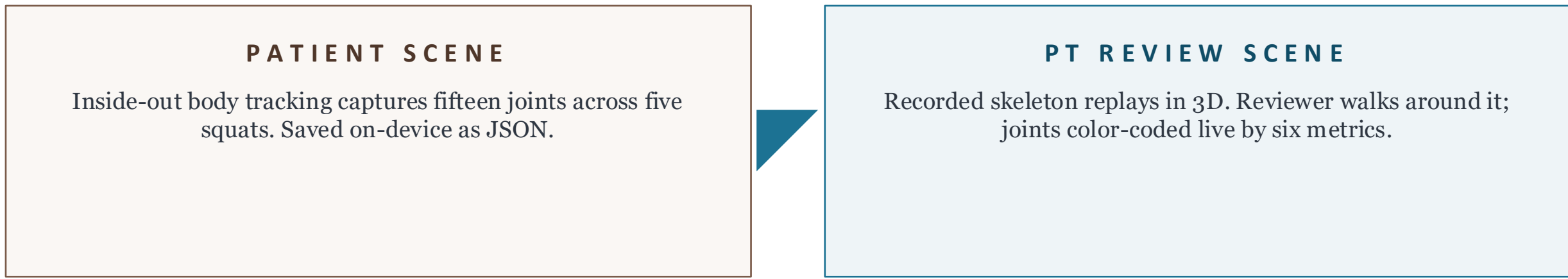


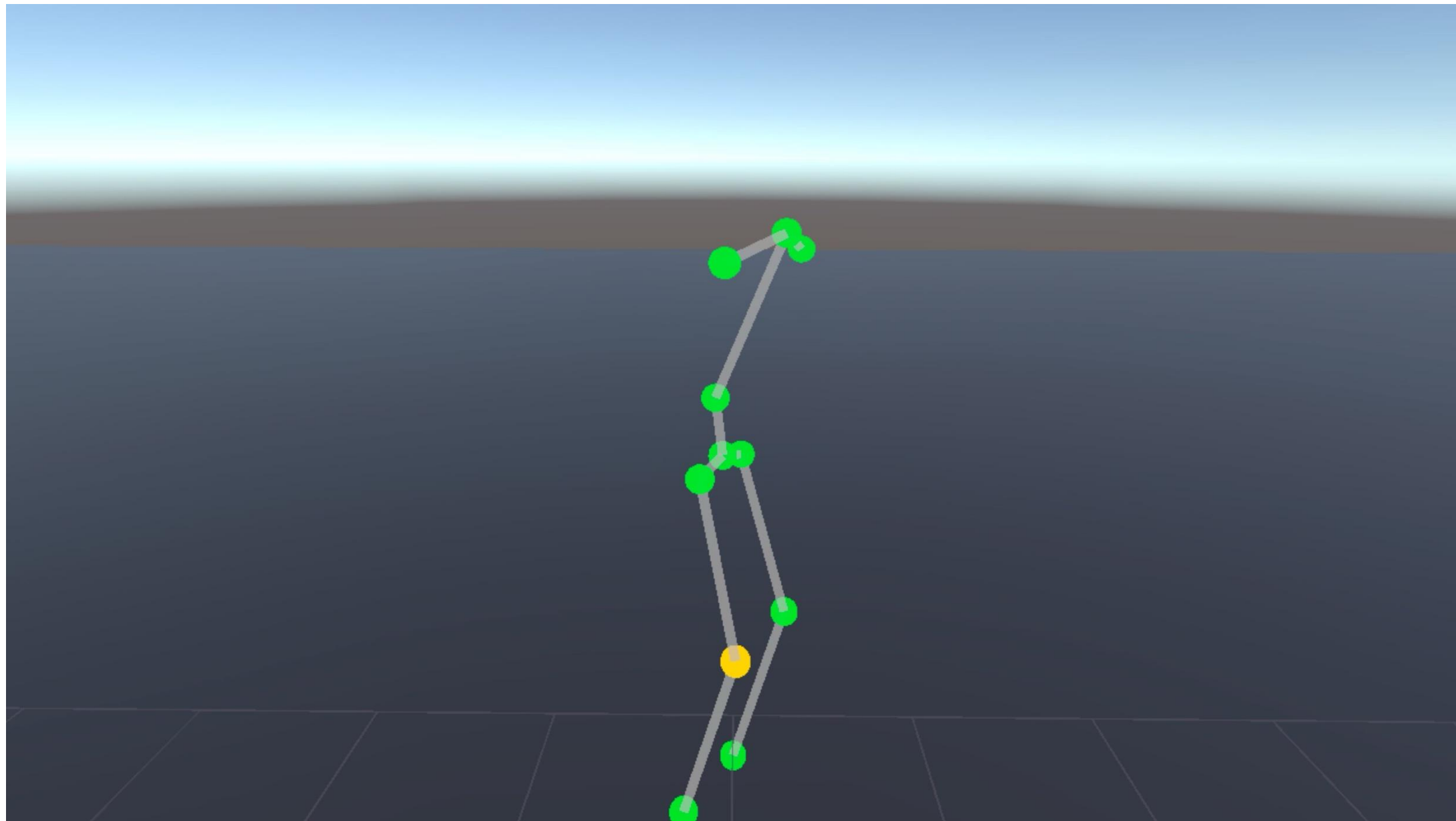
Figure 1. System architecture. The two Unity scenes share recordings via on-device JSON; analysis runs on load before playback.

Table 1. Six biomechanical metrics with literature-backed thresholds.

Metric	Mild / Severe	Source
Knee asymmetry (L–R angle)	6° / 12°	Nae 2017
Hip asymmetry (L–R angle)	6° / 12°	Nae 2017
Knee FPPA (dynamic valgus)	6° / 10°	Bishop 2017
Trunk lateral lean	5° / 12°	Sahabuddin
Trunk forward lean	30° / 45°	Sahabuddin
Limb symmetry index (LSI)	10% / 20%	Wellsandt 2018

Severity color coding

- Normal — within literature range
- Mild — first threshold crossed
- Severe — 1.5× scale + outline



STUDY METHODS

An in-class within-subjects counterbalanced crossover study was conducted with eight participants. Each reviewed the same recorded squat in both VR (PT Detective) and 2D side-view video. Order was counterbalanced: four participants reviewed in VR first, four in 2D first.

For each of four anomaly categories — knee valgus, left–right asymmetry, forward lean, and lateral lean — participants reported whether they observed it (Yes / Maybe / No) and rated their confidence on a 1–5 scale. Six NASA-TLX subscales captured perceived workload, and a forced binary preference question concluded each session. Statistical tests were planned in advance: Wilcoxon signed-rank for paired ordinal scales, two-sided binomial test for the preference count.

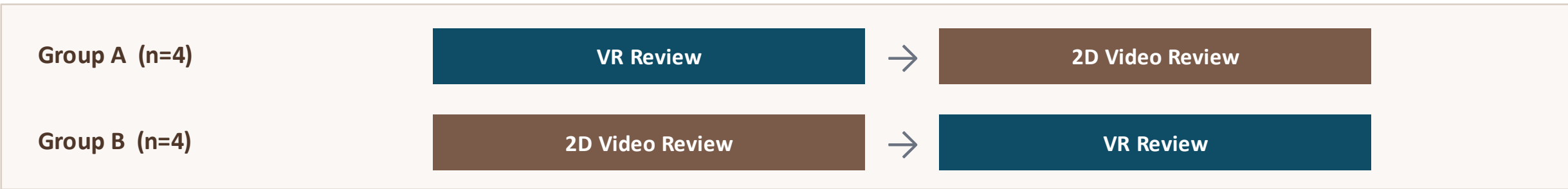


Figure 2. Counterbalanced within-subjects crossover. Each participant reviewed the same recorded squat in both modalities; order was randomized across pairs.

RESULTS

All eight participants chose VR over 2D for movement review (Figure 3; binomial test against a 50/50 null, p = 0.008). Three measures showed very large effects (matched-pairs rank-biserial $|r| \approx 0.91$; Table 2): VR participants reported less perceived task failure, less frustration, and higher confidence detecting left–right asymmetry. Detection counts diverged sharply by anomaly type (Table 3). Asymmetry was the cleanest VR win, with four participants reporting it only after viewing the VR replay; forward lean was the lone 2D advantage, reflecting the natural strength of the video at capturing leaning motion.

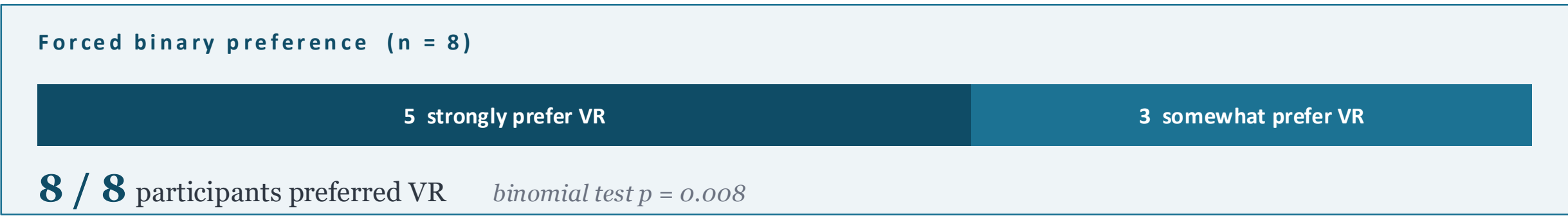


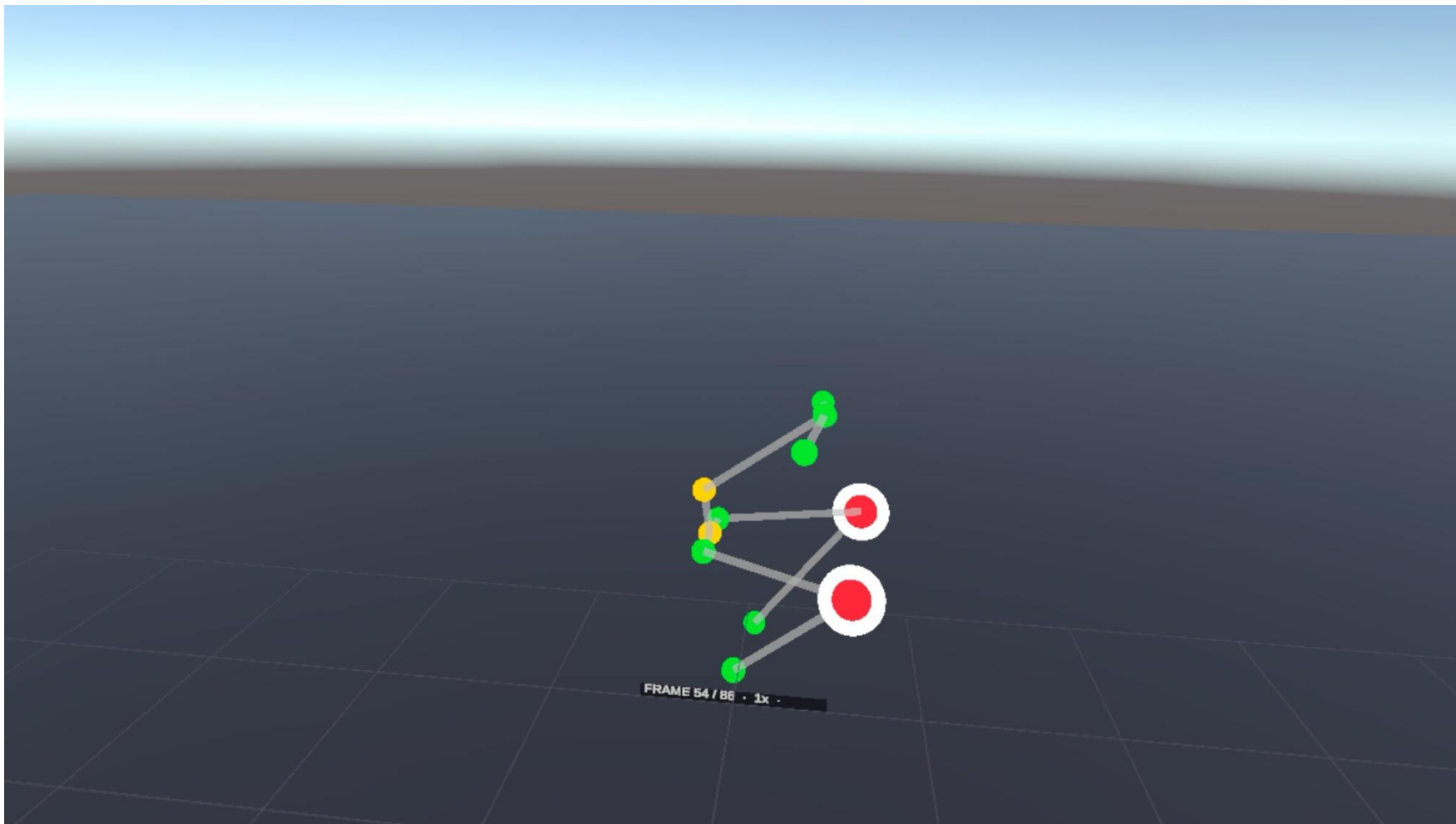
Figure 3. Forced binary preference. All eight participants preferred VR; binomial test against 50/50 yields p = 0.008.

Table 2. Selected outcomes with Wilcoxon signed-rank tests and matched-pairs rank-biserial effect sizes.

Outcome	VR mean	2D mean	r	p
TLX-Performance (lower = less failure)	2.50	4.62	−0.91	0.062
TLX-Frustration	1.75	3.12	−0.91	0.109
Confidence: detecting asymmetry	3.88	2.88	+0.91	0.094
TLX-Physical demand	2.75	1.38	+0.91	0.062

Table 3. Anomaly detection counts (Yes or Maybe, out of 8 participants).

Anomaly	VR	2D	Δ
Left–right asymmetry	5	2	VR +3
Knee valgus	3	1	VR +2
Lateral trunk lean	1	0	VR +1
Forward trunk lean	4	7	2D +3



DISCUSSION

Despite VR scoring higher on physical demand (mean 2.75 vs. 1.38 in 2D), the unanimous preference for VR suggests the visualization paradigm — three-dimensional rotation around the body, with severity-encoded color — provides information that flat video cannot, particularly for laterally-oriented anomalies such as left–right asymmetry and dynamic valgus.

The marginal p-values for the three large-effect findings reflect a structural property of the Wilcoxon test at n = 8: even when every participant moves in the same direction, the test’s discrete-distribution floor lies near p = 0.039. Effect sizes therefore communicate the magnitude of these results more clearly than significance values alone in this regime [4].

The lean advantage for 2D — the only outcome where 2D outperformed VR — is consistent with the natural strength of a camera at capturing sagittal-plane motion. This pattern argues for hybrid review tools that combine 3D VR replay with a complementary side-view video feed, rather than displacing 2D review entirely.

FUTURE WORK

Open research question. Does a hybrid VR + side-view-video review tool — combining 3D rotation and color-coded severity with the sagittal-plane sensitivity of 2D — outperform either modality alone for clinician reviewers? How will improved body tracking affect works?

SELECTED PARTICIPANT FEEDBACK

- “The bright colors made it very easy to identify anomalies.”
— Participant 1B · VR condition
- “Even though the form looks good on the video, it's pretty deceiving in terms of identifying any anomalies in the form.”
— Participant 1B · 2D condition
- “It was a lot easier to identify the individual joints and move around in real space to see how the joints are moving in 3D.”
— Participant 3B · VR condition

ACKNOWLEDGEMENTS

Thank you to course instructors Aarav and David for their guidance and feedback throughout the semester; to the eight study participants for their time and candor; and to the CSCI 1951T cohort for piloting the protocol.

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